

# Kaustubh Rathi

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## SUMMARY

Amazon SDE Intern shipping production backend services and AI-powered operational tooling for the Grocery Ordering organization; final-year B.Tech (IT) at IIIT Allahabad. Codeforces Expert and LeetCode Knight with strong data-structures, algorithms, and distributed-systems fundamentals; experienced in Java, Python, and C++.

## EDUCATION

**B.Tech in Information Technology (Minor in Economics)**

2022 – 2026

Indian Institute of Information Technology (IIIT), Allahabad — CGPA: 8.52/10

Prayagraj, India

## TECHNICAL SKILLS

**Languages:** Java, Python, C++, SQL

**Backend & Distributed:** Coral (RPC), Smithy, REST APIs, microservices, concurrency, WebSocket, fault tolerance, observability

**Cloud & Tools:** AWS (Lambda, DynamoDB, API Gateway, CloudWatch, Step Functions, Bedrock, CDK), Docker, Git, MySQL, MongoDB

**AI/ML:** PyTorch, TensorFlow, LangChain, RAG, NER, scikit-learn

## EXPERIENCE

**Software Development Engineer Intern**

Jan 2026 – Present

Amazon — Grocery Ordering, Stores

Bengaluru, India

- Architected a Java Coral microservice (AWS Lambda + API Gateway, Smithy-modeled) over a DynamoDB layer with 3 GSIs tuned to primary access patterns and a generic reusable DAO with Dagger DI; authored its CDK infrastructure (DynamoDB stacks, cross-account IAM, VPC endpoints) powering the grocery-ordering agent pipeline.
- Engineered an autonomous order-failure remediation pipeline: an LLM agent (Amazon Bedrock) investigates ~150 failed orders/day, semantically clusters them by root cause, and auto-generates categorization-rule code changes (CRs) to prevent recurrence — orchestrated by a deterministic, crash-resilient state machine with per-call output validation, retries, and human approval gates.
- Shipped a 24/7 incident-investigation agent end to end that scans tickets every 5 minutes and correlates CloudWatch, Lambda, and Step Functions evidence with recent deployments to auto-generate 5-level root-cause analyses, cutting investigation time from ~2 hours to ~15 minutes.

**Artificial Intelligence Intern**

Dec 2024 – Feb 2025

ASPR Agrovalue (I) Limited

India

- Automated an invoice-processing pipeline using named-entity recognition (NER) to extract structured fields (item, quantity, unit price) from 10,000+ supplier invoices/month into a normalized schema, replacing manual data entry.
- Developed a Python web-scraping + ETL pipeline aggregating 1,000+ market and price records/day from fragmented sources into a clean database, with scheduled incremental refresh and data-quality checks.
- Designed an internal Streamlit dashboard surfacing commodity price trends and data-quality views, adopted by the procurement team in place of manual spreadsheet reports.

## PROJECTS

**Deep Researcher** | Python, LangGraph, FastAPI, Flask, WebSocket, FAISS

🔗 GitHub

- Built a multi-agent research-automation web app orchestrating 7 specialized LLM agents (research, edit, write, review, revise, publish) with a FastAPI backend and WebSocket log streaming to the frontend; exports reports to PDF/DOCX/Markdown.

**Black-Scholes Option Pricer** | Python, Streamlit, py\_vollib, SciPy

🔗 GitHub

- Deployed a Streamlit tool that fetches live market prices and computes option prices and Greeks (delta, gamma, vega, theta) with dividend-yield adjustment.

**Medical Chatbot** | LangChain, FAISS, OpenAI, Gradio, Docker

🔗 GitHub

- Created a Dockerized RAG clinical-interview chatbot supporting 6 languages (including Hindi) that conducts structured interviews and generates assessment reports.

## ACHIEVEMENTS

- **Codeforces Expert** (max rating 1751) | **LeetCode Knight** (rating 2000+), 800+ problems solved.
- Secured **99.45 percentile** in JEE Mains 2022 (top 0.55% among 1M+ candidates).
- Completed *Options 101* by Akuna Capital (derivatives and options trading fundamentals).

## LEADERSHIP

**Overall Coordinator, Robita Club (Generative AI), IIIT Allahabad (2023–25):** led 10+ students and mentored 100+ juniors through campus-wide AI hackathons and PyTorch workshops.